Finding primary research

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which inspected document and locator support this exact statement.

Teach primary evidence without assuming originality guarantees quality or inventing unavailable text.

Keep one habit: Model conflicting counts → separate versions → inspect scoped original result → remove unavailable card → assess dataset mismatch record.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of finding primary research

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of reading a research claim

Reference example, not the only acceptable wording: Teacher explanation: Reading a claim requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Separate the claim from supplied **evidence**. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Pictured end goal

Explain finding primary research using the records and boundaries of this fictional investigation.

<details>
<summary>▼ Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Reading a research claim

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Primary research presents a study's own methods and results. A summary can help discovery but should not replace inspection of the original **evidence**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.
</details>

Outcomes

Explain the **mechanism** and terminology of finding primary research

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach source identity, version and access as part of claim verification

> Do not turn fictional cards or missing original text into real-source verification.

Definitions and complete source-record contract

Primary research presents a study's own methods and results. A summary can help discovery but should not replace inspection of the original **evidence**.

Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version.

Source-tracing requirement	Complete classroom meaning
Primary report	Study authors describe their own methods/results; originality alone does not establish quality
Secondary summary	Blog/search summary describes another report; useful discovery lead

Source-tracing requirement	Complete classroom meaning
Source record fields	Title, authors, version, date, identifier, access location and claim locator
Claim context	Dataset, metric, denominator, method and exact scoped result
Version discipline	Keep original/revised records distinct; do not silently mix their counts or methods
Code and data	Separate artifacts with own versions/identifiers; neither automatically substitutes for the paper
Unavailable original	Record access gap and secondary status; no direct-primary verification claim
Exercise status	All document cards and identifiers below are fictional internal labels, not real papers/DOIs/URLs
Missing original data	A reported count can be attributed to its version without claiming an independent raw-data audit

Main document-card ledger and checked reasoning

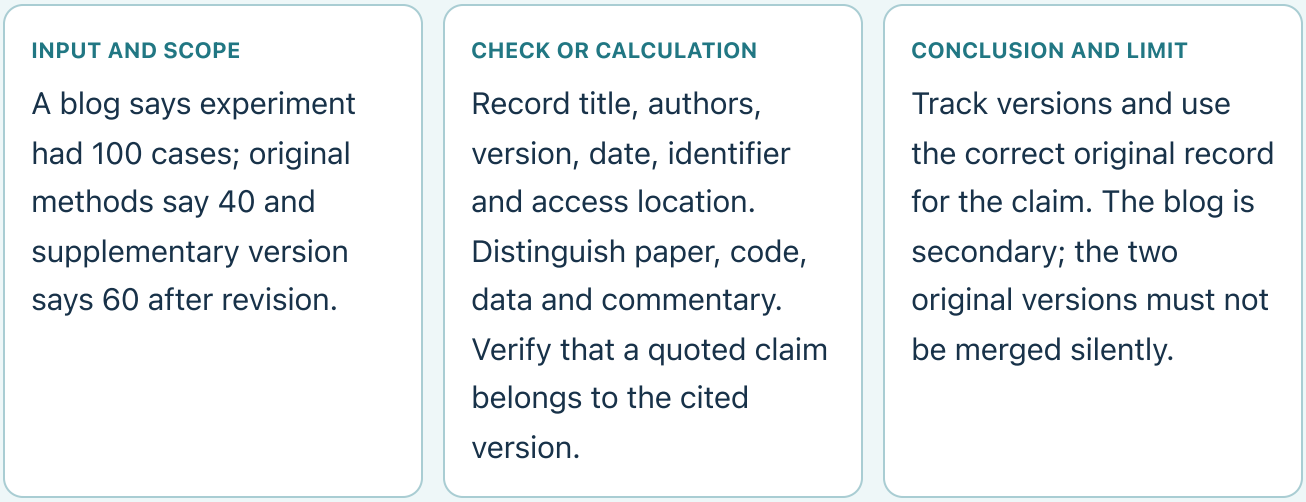
Added fictional document card	Identity and access location	Supplied claim locator/content
Blog summary	Title “Room retrieval study overview”; author Fictional Newsletter Team; v1; classroom date 2026-01-03; ID BLOG-R; this table/card BLOG-R	Summary paragraph: study used 100 cases; no supporting primary locator supplied
Original study record	Title “Room Retrieval Pilot”; author Fictional Team R; v1; classroom date 2026-01-01; ID STUDY-R-v1; this table/card STUDY-R-v1	Methods §2: 40 constructed cases; raw case appendix not supplied in this lesson
Revised supplementary record	Title “Room Retrieval Pilot — revised supplement”; author Fictional Team R; v2; classroom date 2026-01-02; ID STUDY-R-v2; this table/card STUDY-R-v2	Revised methods §2: 60 constructed cases; reason/membership relation to v1 not supplied
Faithful versioned record	STUDY-R-v1 §2 → 40; STUDY-R-v2 §2 → 60	Blog’s 100 is unsupported by these supplied originals; do not add counts or infer disjoint cases

Added fictional document card	Identity and access location	Supplied claim locator/content
Remaining evidence gap	Raw case identities and revision explanation unavailable here	Attribution verified within supplied cards only; no independent raw-data or real-literature verification

Main BLOG-R 100 is unsupported by supplied original v1 methods 40 and revised v2 methods 60. Keep exact version/date/locator; do not add 40+60, infer disjointness or invent revision rationale/raw data. Added identities/cards are explicit classroom fiction, not actual research citations. Picture BENCH-v1 Table 1 records 9/10 correct on fixed labelled tasks, not general 90% reliability.

If original unavailable, label secondary account and primary verification not performed. Originality does not guarantee method quality. Selected summary/paper different-dataset case: record exact paper/version, actual dataset, metric and reported result with locator; mark search-summary mismatch and do not borrow its score.

Finding primary research — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Qualification: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision.

Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Version, artifact and access probes

Source-tracing probe	Expected interpretation
Blog and primary count disagree	Record mismatch; cite the original version supporting the statement
Original40, revised 60	Keep version-specific counts; no silent sum, average or replacement
Version date is newer	Does not make every earlier claim invalid; inspect what changed
Paper identifier known but text unavailable	Bibliographic discovery, not inspected methods/results
Only code or dataset found	Record separate artifact role/version; paper interpretation still needs evidence
Primary paper contains a claim	Still inspect method, metric and limitations; original is not automatically correct
Quote has no page/section/table locator	Add the claim’s exact accessible location

Source-tracing probe	Expected interpretation
Search summary uses different dataset	Do not borrow its number into the linked-paper comparison
Fictional practice cards completed	No real literature search, primary-paper access or raw-data audit claimed

Separate benchmark report identity and scoped result

Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report.

Blog says "90% reliable". Original report records 9/10 correct on one fixed fictional task set and describes its labels.

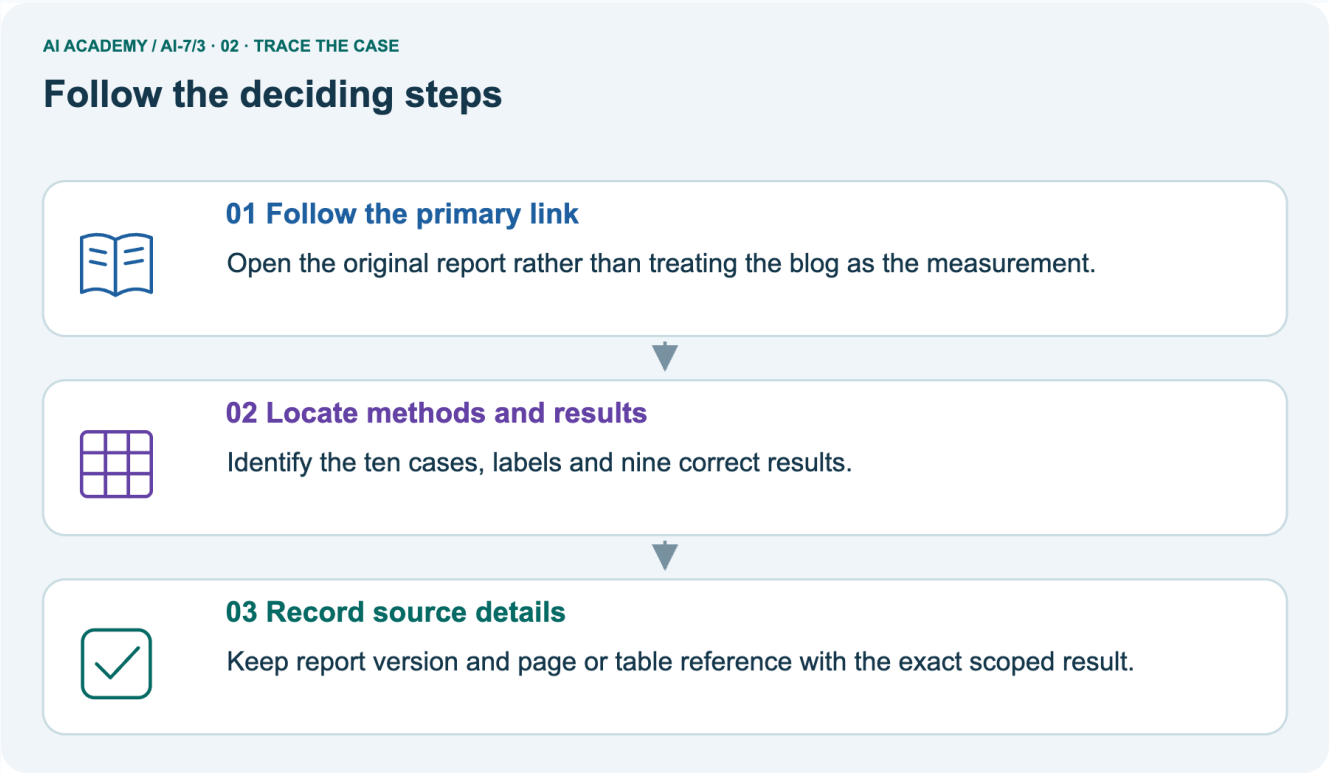
The picture is a separate fictional ten-case report. Its blog says “90% reliable”; the original records 9/10 correct on one fixed task set and describes the labels. A faithful claim is “the report records nine correct out of ten on its stated cases, ” with exact version and table reference, not general reliability.

For this practice, use fictional report ID BENCH-v1, version 1, results Table 1 and methods §2 as supplied locators, not a real citation. If that original is unavailable, keep the blog claim marked secondary and primary result unverified. Do not say you opened or inspected missing methods, and do not infer raw records from the percentage alone.

Separate fictional picture card	Supplied report detail
Identity	Title “Fixed Task Benchmark Note”; author Fictional Team B; version 1; classroom date 2026-02-01; ID BENCH-v1
Location	This supplied picture/card; methods §2 and results Table 1 are fictional locators
Methods §2	Ten fixed fictional tasks with supplied correct-answer labels
Results Table 1	Nine correct / ten tasks = 90% on this task set
Secondary blog	Headline “90% reliable” omits task-set scope

Separate fictional picture card	Supplied report detail
Unavailable-original variant	Only secondary headline accessible; methods/results verification not performed

Picture trace and unavailable-original key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Follow the primary link	Open the original report rather than treating the blog as the measurement.
2. Locate methods and results	Identify the ten cases, labels and nine correct results.
3. Record source details	Keep report version and page or table reference with the exact scoped result.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

This example names fictional documents; it is a source-tracing exercise, not a real literature citation.

Change: The original report is unavailable and only the blog remains.

Prediction: Mark the measurement as unverified rather than filling in absent methods.

Reason: The summary cannot supply details that have not been inspected.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Primary source	A record reporting its own study
Version identifier	A label distinguishing revisions
Secondary source	A summary or interpretation of another source

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Finding primary research

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Finding primary research. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Separate what the blog claims from what the original report records. Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

Open the original report rather than treating the blog as the measurement. Identify the ten cases, labels and nine correct results. Keep report version and page or table reference with the exact scoped result.

Changed case: Mark the measurement as unverified rather than filling in absent methods. The summary cannot supply details that have not been inspected.

Changed-case reasoning: Mark the measurement as unverified rather than filling in absent methods. The summary cannot supply details that have not been inspected.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main supplied originals report 40 and 60 in distinct versions; blog 100 remains unsupported and counts must not merge. Picture original 9/10 supports that fixed-task result only, while unavailable-original variant remains secondary/unverified. Selected search-summary mismatch needs exact actual paper/version, dataset, metric, result and locator; record discrepancy rather than borrowing its number.

All exercise identities are fictional and no raw-data or live-literature verification is claimed.

Diagnose the current misconceptions

The worked result establishes every future case: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

Teacher explanation: Exact paper/version, dataset, metric and reported result; mark the search-summary mismatch rather than borrowing its number.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Primary research presents a study's own methods and results. A summary can help discovery but should not replace inspection of the original **evidence**.

Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version.

W2 Trace the worked case

A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

W3 Audit the limit

A report omits this qualification: Being primary does not guarantee soundness; methods and **evidence** still require assessment. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Being primary does not guarantee soundness; methods and **evidence** still require assessment.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Qualification: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

Teacher explanation: Exact paper/version, dataset, metric and reported result; mark the search-summary mismatch rather than borrowing its number.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Exact paper/version, dataset, metric and reported result; mark the search-summary mismatch rather than borrowing its number. Require intermediate reasoning and the stated scope.

Fresh retry: Repository README differs from paper v1, with a later release note. Expected reasoning: Record code release and paper version separately; investigate which experiment each number refers to.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled finding primary research **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of finding primary research.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Methods results and limitations. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/4, ai-7/6, ai-7/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of finding primary research

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Check every learner before discussion: Use the worked case for Finding primary research. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Being primary does not guarantee soundness; methods and **evidence** still require assessment.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The original report is unavailable and only the blog remains. Require a **prediction** and a reason before comparing. Expected result: Mark the measurement as unverified rather than filling in absent methods.

Reason: The summary cannot supply details that have not been inspected.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Finding primary research: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

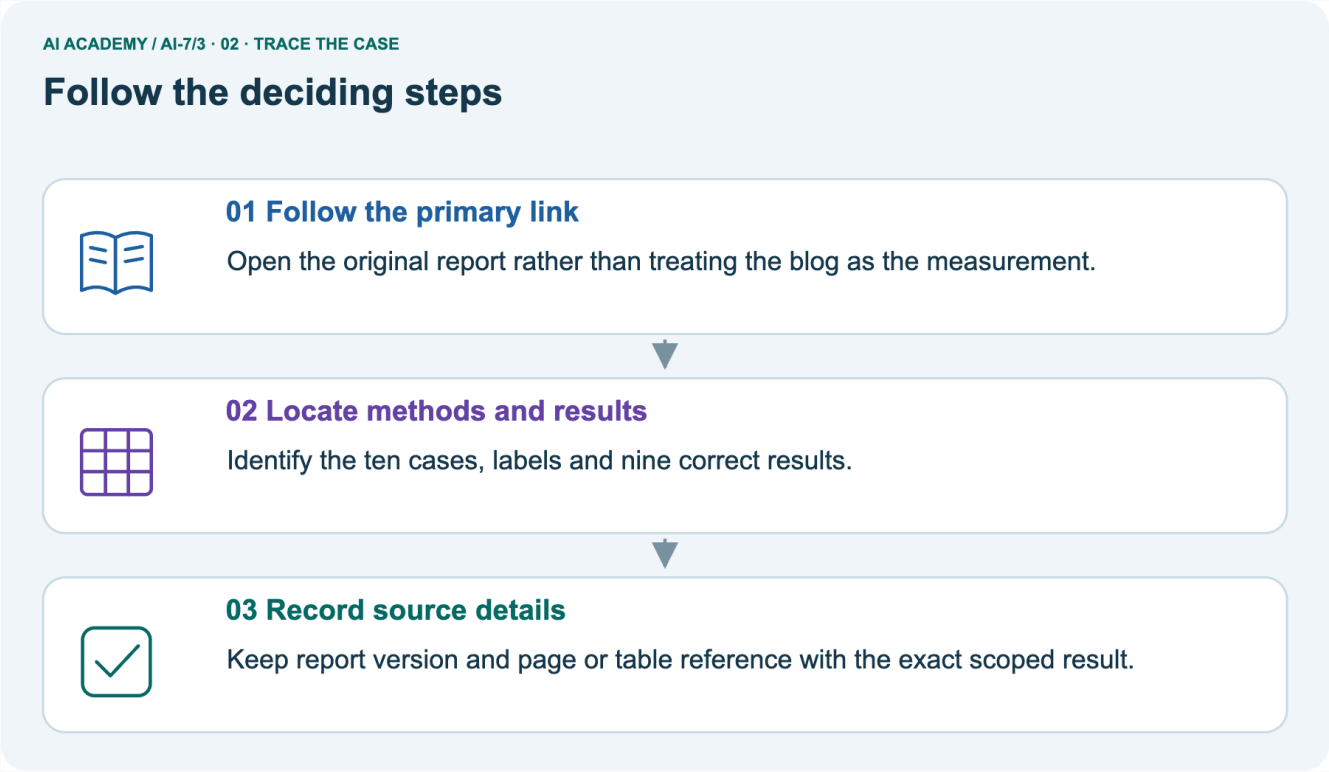
Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report. Supplied facts: Blog says "90% reliable".

Original report records 9/10 correct on one fixed fictional task set and describes its labels.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



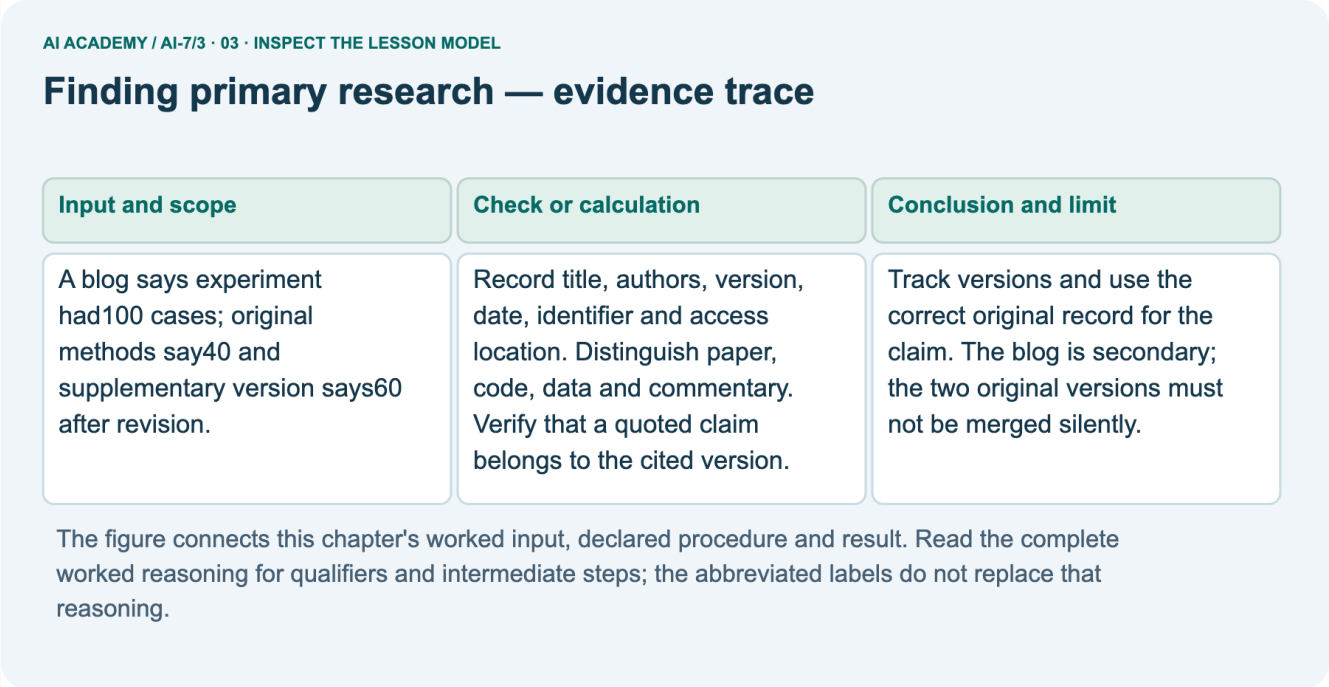
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the primary link: Open the original report rather than treating the blog as the measurement. → Locate methods and results: Identify the ten cases, labels and nine correct results. → Record source details: Keep report version and page or table reference with the exact scoped result.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Finding primary research — evidence trace



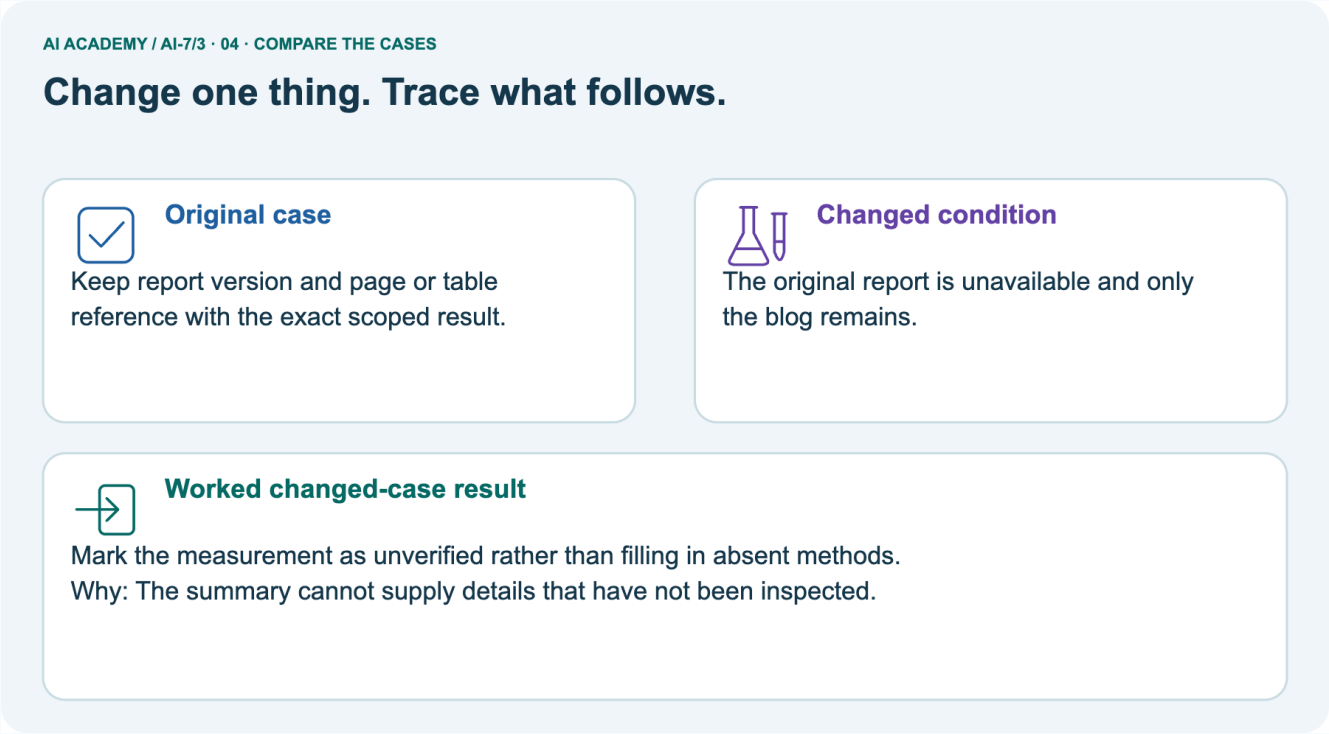
Follow the labelled picture with the matching explanation. Open the vector diagram

Finding primary research — **evidence** trace. A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. | Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version. | Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep report version and page or table reference with the exact scoped result.
Changed condition: The original report is unavailable and only the blog remains.

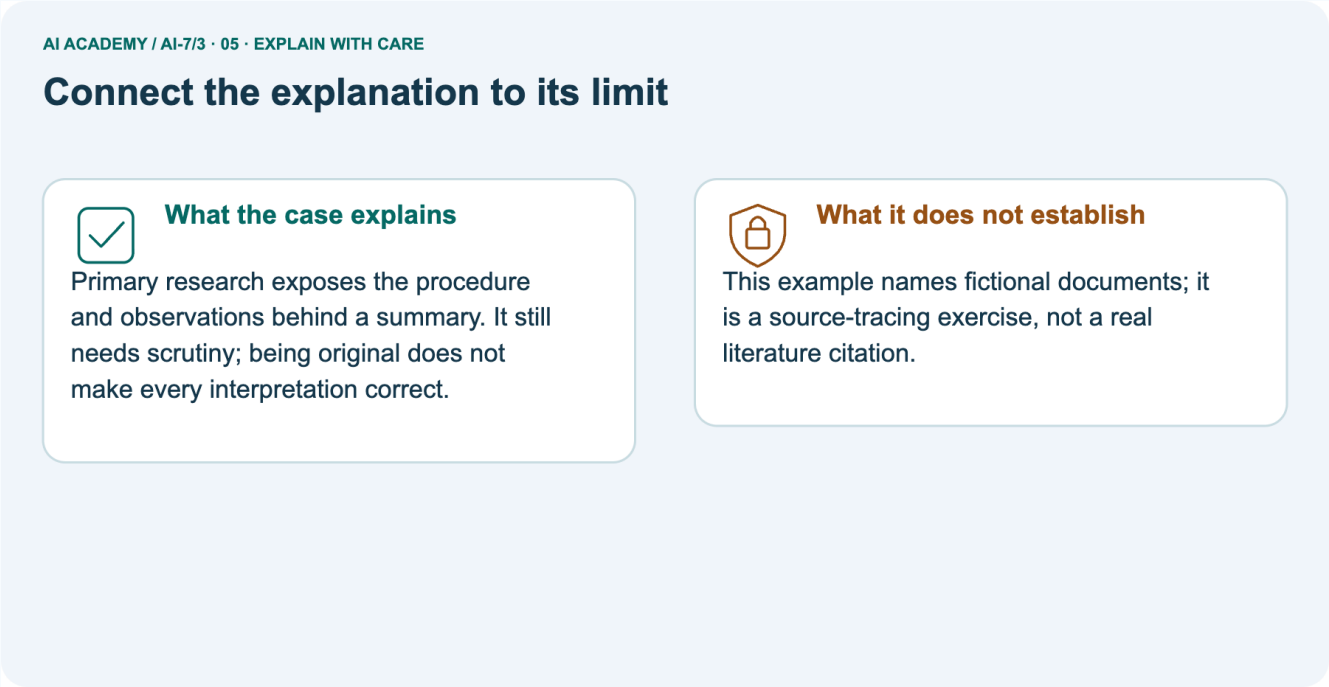
Expected result: Mark the measurement as unverified rather than filling in absent methods.
Reason: The summary cannot supply details that have not been inspected.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The summary cannot supply details that have not been inspected.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

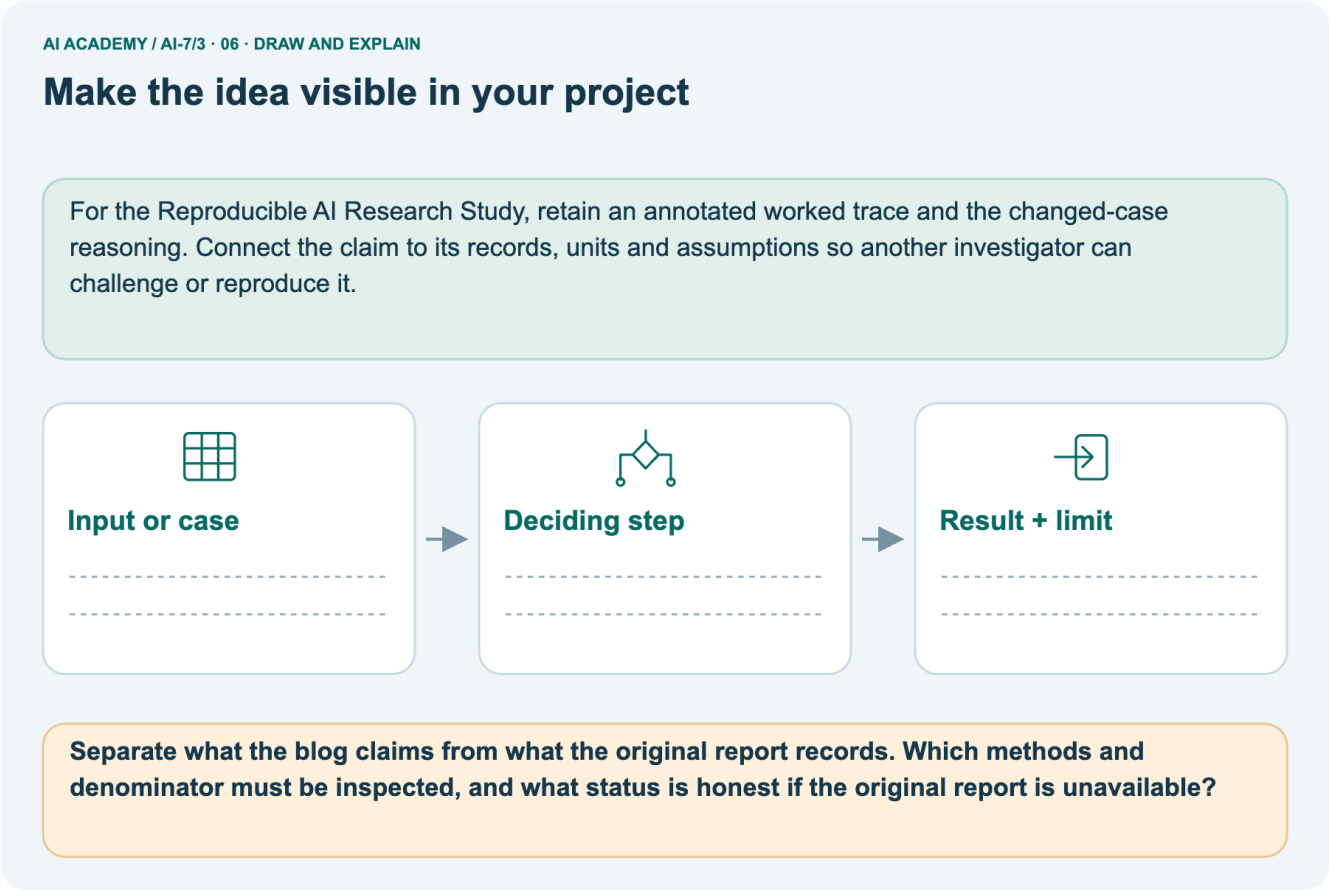
Limit: This example names fictional documents; it is a source-tracing exercise, not a real literature citation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This example names fictional documents; it is a source-tracing exercise, not a real literature citation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Open the original report rather than treating the blog as the measurement. Identify the ten cases, labels and nine correct results.

Keep report version and page or table reference with the exact scoped result. Changed case: Mark the measurement as unverified rather than filling in absent methods.

The summary cannot supply details that have not been inspected.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Finding primary research: the case

02. Follow the deciding steps

03. Finding primary research — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.